Contourlet Domain Feature Extraction for Image Content Authentication

Ali Bouzidi and Nadia Baaziz
Department of Computer Science and Engineering
Université du Québec en Outaouais
Gatineau, Québec, Canada
boua04@uqo.ca; nadia.baaziz@uqo.ca

Abstract— **The achievement of multimedia content authentication by means of digital watermarking, while never easy, is further complicated by the continuing investigation of different ways of generating authentication signatures which survive specific acceptable manipulations. In this paper, we present a novel approach which takes advantage of a multiscale framework and directionality to extract the significant features of an image from its redundant contourlet transform. The introduced redundancy brings simplicity and accuracy for feature calculation. We also describe the applied post-processing steps with the aim of stabilizing those features under acceptable image manipulations, namely lossy JPEG compression and additive noise corruption. Many experiments and comparative studies are performed to show the effectiveness of our technique in generating content signatures based on invariant image features, as well as to demonstrate its superiority when compared with a redundant wavelet approach.**

Keywords— image authentication; invariant feature; contourlet transform.

I. Introduction

Since the growth of network and the widespread use of the Internet, the exchange and the reproduce of multimedia become easier and consequently increase the fraud. In fact, many researches have been made to protect, locate and report any malicious manipulations. However, to provide security services, mainly the use of content-based authentication techniques seems a solution to trade-off between acceptable manipulations [7] and malicious manipulations. Such lossy compression, enhancement, rotation, scaling, light additive noise, and watermarking are examples of acceptable image manipulations since visual content is preserved and human visual perception and understanding of the image are not affected. In other words, ideal image content authentication should be able to detect any malicious attack which affected the image in a significant manner such as adding, erasing or replacing object into the original image.

One possible approach to content authentication by means of image watermarking is to design a watermark as one or many digital signatures of the image content. Therefore, the watermarking process has to satisfy the following requirements: 1) the watermark has to depend on some essential image features that would approximately stay invariant under acceptable manipulations, 2) the watermark construction may undergo various data transformations such as key-dependent random processing in order to enhance security against potential attacks (guessing and forgery), and 3) the watermarking scheme must use robust embedding (difficulty to remove the mark without introducing severe distortions into the image) and must allow for accurate watermark recovery without using the original unmarked image [5]. Then, authenticity detection for a given image simply compares its feature-based signatures with those extracted from the recovered watermark. The differences, if any, indicate the presence of malicious alterations and their location; the image is otherwise declared to be authentic.

An example of image content authentication framework combining invariant feature-based signature with Error Control Coding and Public Key Infrastructure is described in [8]. This framework is interesting as it offers the advantage of being generic and expandable to various feature extraction methods. The quality of the selected features relies mainly on the degree of the invariance property that is ensured under acceptable manipulations and significantly affected for malicious alterations. Examples of features used to provide image authentication include edges, colors, textures, gradient, luminance, transform coefficients, etc. Lin and Chang [4] propose a feature extraction procedure based on the invariance of the relationships between two discrete cosine transform coefficients at the same position in a block pair before and after JPEG compression. In [9], wavelet-based adaptive denoising and morphological operations provide significant invariant features used for signature generation. Many studies in the area of robust visual hashing for search and indexing in large image databases provide useful and powerful tools for invariant feature extraction. Image hashing schemes are constructed by combining many ideas from the fields of signal processing, error-correcting codes, and cryptography [6,10,11]. In the Venkatesan's scheme [11], the mean or the variance of blocks are calculated from a random tiling of the wavelet subbands. The resulting feature vector is quantized and compressed using the decoding stage of an error-correcting code. In [6], the authors propose an iterative filtering scheme with repeated thresholding to extract main geometric features within an image. A randomizing process based on a secret key is then applied to achieve image-hash requirements. The algorithms described in [6] and [11] produce compact image hashes that approximately stay invariant under additive noise and common image filtering operations. In their recent work, Swaminathan et al. [10] perform a weighted summation of the discrete polar Fourier transform over random subsets. The derived image features achieve resiliency against common filtering operations as well as geometric transformations.

In this paper, we describe a novel scheme for image feature extraction operating in the redundant contourlet domain. Our proposed approach exploits a multiscale framework to generate binary maps that capture significant image features across spatial and directional resolutions. The practicality and accuracy of this approach are augmented with the introduced spatial redundancy through oversampling, yielding equal-size contourlet subbands [1]. The paper is organized as follows. In the next section, we discuss the main characteristics of the contourlet transform and describe our construction of redundant contourlet decomposition. Section III highlights the formulation and computation of the directional feature maps followed by the process of compaction and stabilization. In section IV, we present our experimental results and we compare them against the results obtained with a redundant wavelet-based approach. Then we assess the algorithm performance in terms of feature invariance under acceptable and malicious manipulations.

II. THE REDUNDANT CONTOURLET TRANSFORM

The contourlet representation of an image is constructed by applying two successive decomposition stages [3]. The first stage transforms the original image into a Laplacian pyramid (LP) having *L+1* scale levels. The second stage is a decomposition of each LP scale level into D subbands through a directional filter bank structure using quincunx filters and critical down-sampling operators. The overall result is a directional multiscale image decomposition based on sparse and oriented representation of smooth edges. Some preliminary studies on the contourlets [3, 1] indicate great potentials in image analysis domain and provide enough motivation to exploit such image transform in extracting and representing significant image features.

Even with the attractive properties, there is still a major concern with the sampling rate which is leading to a variation in the subband sizes across the scales. A redundant contourlet transform was introduced in our work to allow for practical and accurate processing on equal-sized directional subband images [1]. The redundancy is achieved by discarding any down-sampling operation in the Laplacian pyramid scheme. The end result is a redundant Laplacian pyramid (RLP) with *L+1* equal-size scale levels: one coarse image approximation and *L* bandpass images. The redundancy factor is up to *L*. By applying the same *D*-level directional decomposition (with critical sampling) on each RLP bandpass, we obtain a redundant contourlet transform with *LD* equal-size directional subbands in addition to the coarse image approximation (see Fig. 1). Each contourlet subband is represented by a sub-image $\{C_{l\,d}(i, j), i=1..M, j=1..N\}$, where l is its scale level, d is its frequency direction and MN is the total number of its coefficients. The size of the coarse image approximation C_L is $2M\text{x}2N$ coefficients.

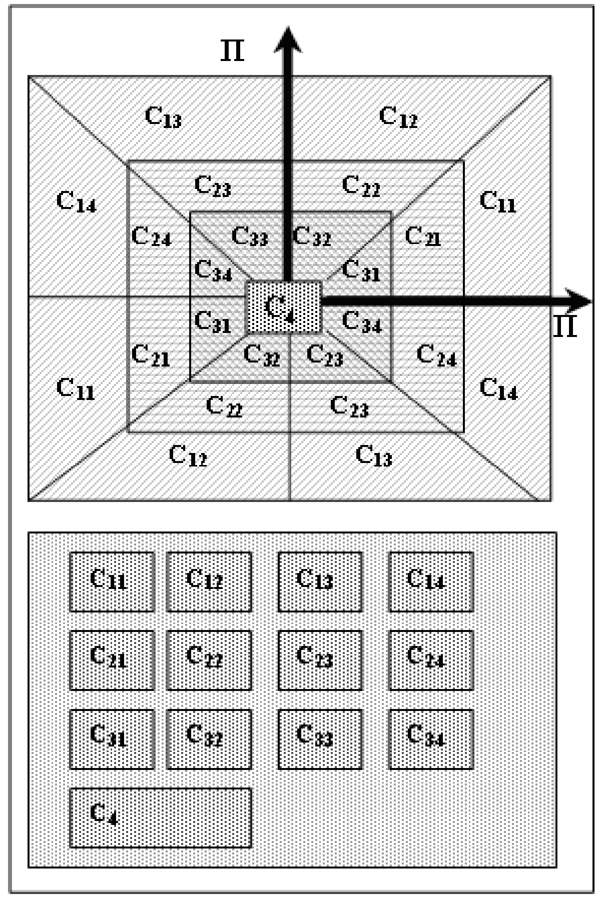

Fig. 1 (top) Frequency and (bottom) spatial representations of a redundant contourlet decomposition with *3* scale levels (*L=3*) and *4* frequency directions (*D=4*). The coarse image approximation (*C4*) is not decomposed directionally.

III. SIGNIFICANT FEATURE EXTRACTION

Basic processing consists in first extracting image features in the contourlet domain and then applying selection and stabilizing processing to gather significant image features and achieve a high degree of invariance under acceptable manipulations.

A. Contourlet Feature Extraction

The feature extraction procedure expands on a well-known method originally developed to capture perceptually significant image characteristics across the scales of a wavelet transform [2]. Adaptation to the redundant contourlet domain is accompanied by modifications. We have mainly introduced directional processing across multiresolution directional subbands and exploited the transform's redundancy to improve accuracy and localization of captured contourlet features. For each frequency direction d of the contourlet decomposition, contourlet features are extracted through the calculation of *a* directional map M_d, as formulated in the following equation:

$$M_d(i,j) = \frac{1}{2} B(i,j)\, T(i,j)^{0.2}\; E(d,i,j)^{0.2} \quad (1)$$

where

$$B(i,j) = 1 + \frac{1}{256} C_{L+1}(2i-1, 2j-1) \quad (2)$$

$$T(i,j) = Var\left\{ C_{L+1}(2i-1+y, 2j-1+x) \right\}_{x=0,2;\, y=0,2} \quad (3)$$

$$E(d,i,j) = \frac{1}{4}\sum_{l=1}^{L}\sum_{x=0,1}\sum_{y=0,1}\left[C_{l,d}(i+y, j+x)\right]^2 \quad (4)$$

The factor B is proportional to the brightness values of the coarse resolution sub-image, whereas T and E capture, respectively, the local texture in the lowpass subband and the presence of oriented image edges across the scales of directional resolutions.

Note that $C_{l,d}$ is a contourlet subband at a scale level l and a frequency direction d, while C_{L+1} is the coarse scale sub-image (or image approximation). The overall result of this processing is a set of D contourlet feature maps.

B. Significant Features and Stabilization

Many post-processing steps are applied to the extracted contourlet features (in section A) aimed at selecting significant image features and stabilizing their values against specific acceptable manipulations, namely JPEG compression and additive noise.

The first step is a classification of the contourlet feature coefficients into significant and insignificant coefficients. For this purpose, the coefficients of each directional map are thresholded to obtain binary significant feature maps. The choice of the threshold value has a direct impact on the number of significant coefficients (map density in terms of one values) .

The second step is a directional merge operation which compresses the density of the significant feature maps. This is performed by adding together all of the binary coefficients located on the same positions in all directional maps. The result is a compact binary map of significant image features (SF).

The third step aims at stabilizing the coefficient perturbation caused by acceptable image manipulations. For this purpose, smoothing and clustering of the binary significant coefficients are required. This requirement can be achieved by means of mathematical morphology operations. We have adopted an erosion procedure with a small structuring element (2x2) to get rid of the most isolated coefficients in the significant feature map (SF).

C. Redundant wavelet-based feature extraction

For comparison purposes, and given the popularity of the wavelet transform, we developed a redundant wavelet image representation and adapted the feature extraction algorithm to this new structure. First, the original image of size (*2Mx2N*) is decomposed into 3 directional subbands (H_1, V_1 and D_1) and one image approximation (C_1) using a one-level orthogonal wavelet scheme. The subimage (C_1) is then decomposed into *L-1* levels using an overcomplete (scale-invariant) wavelet scheme. The result is a set of *3L* directional subbands $\{Hi, Vi, Di,\ i=1..L\}$ and one coarse image approximation (C_L). All subbands have the same size *MxN*, and equal the contourlet subband size. The resulting redundant wavelet structure has many analogies with the redundant contourlet structure, but differs in the number of directional resolutions (3 for the wavelets and 4 for the contourlet). Therefore, the efforts invested in correctly structuring the redundant wavelet decomposition led to a straightforward adaptation of the feature extraction technique to wavelets.

IV. Experimental Results and discussion

We have conducted a sequence of tests on various kinds of images and have applied some acceptable image manipulations, mainly JPEG compression with differing quality factors and additive Gaussian noise corruption.

The experiments presented in the following were performed on the images Lena, Boat, Baboon and Peppers, each of size 512x512. Each image was represented as twelve equal-size subbands of size 256x256 and one coarse image approximation of size 512x512. This was the result obtained from a redundant contourlet transform with three scale levels and four directions (L=3 and D=4). Feature extraction (described in section III.A) was then applied to the contourlet structure, yielding four contourlet feature maps, as shown in Fig. 2. Already at this stage, one can notice the potential of our method to gather and localize oriented geometrical patterns in the image. Significant image features are selected through successive post-processing steps such as thresholding, map merging, and coefficient stabilization by morphology operations. More or less map compression can be achieved basically through threshold adjustment during the binarization process, while smoothness and clustering properties are improved through morphology processing.

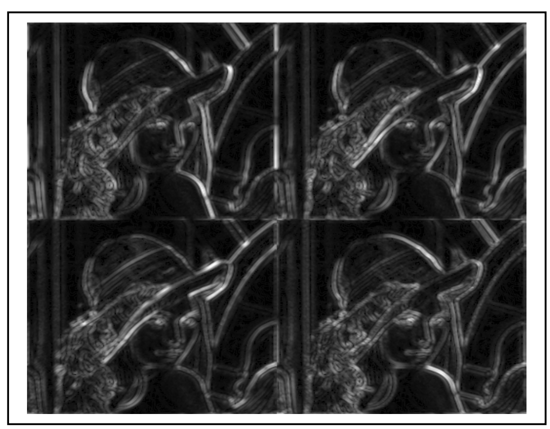

Fig. 2 The 4 contourlet image feature maps, respectively, M_1, M_2, M_3, and M_4 (from top left to bottom right), calculated from a redundant contourlet of the Lena image with $L=3$ and $D=4$.

In the presented experiments, the threshold value has been taken as the median value of the coefficients, and the morphological operation is a 2x2 erosion. The binary map shown in Fig. 3 (left side) is an example of extracted significant features (SF) from Lena image. In order to assess the effectiveness of our method in terms of feature invariance (degree of feature stability), the binary SF map corresponding to an original image was compared to those obtained from manipulated images. Both Hamming distance and Normalized Hamming distance were used to measure the degree of similarity between the compared features. Very low distance values are expected for similar SF maps. Under acceptable manipulations, the SF maps were visually similar, and their absolute difference was examined to appreciate the amount and the location of introduced distortion, thus demonstrating the potential of significant features in surviving such manipulations. In Fig. 3 and Fig. 4, the SF differences exhibit only a few change points yielding sparse difference maps. This is an important property to be preserved under acceptable manipulation in addition to the normalized Hamming distances which are expected to be very low in these situations. Table I provides density measures (number of 1 value) in the original SF maps, in the corrupted SF maps and their differences for the Lena image (or Hamming distance). These results also illustrate the contribution of morphological operations in stabilizing the significant features (see Fig. 4). Fig. 6 and Fig. 7 show the plots of the normalized Hamming distance under a variety of JPEG compressions and additive noise respectively.

As done in contourlets approach, many significant image features were extracted from redundant wavelet structures. To allow for an accurate comparison between the two approaches, the threshold value of feature selection was adjusted in order to achieve the exact map density (number of 1 value) as obtained in the original contourlet significant features. Finally, all comparison results have shown a lack of feature stability with the wavelet approach. Indeed, the introduced feature distortion under acceptable manipulations is always higher than the corresponding contourlet feature (see Table II). Many malicious manipulations have been carried out through adding and removing objects from the original images. These image corruptions resulted in localized and significant changes over the extracted significant feature maps. In these situations, even if the altered objects are very small, the normalized hamming distance values could be low but the difference between original and manipulated SF maps exhibit dense and compact regions as shown in Fig. 8.

TABLE I. MAP DENSITY (NUMBER OF 1 VALUES) OF SIGNIFICANT FEATURES (SF) ILLUSTRATING FEATURE STABILITY UNDER ACCEPTABLE MANIPULATIONS (ADDITIVE NOISE-39.6DB, JPEG-36.3DB-).

Image: Lena	Without morph. Operation	With morph. operation
a) Original SF	3468	2016
b) Noise-corrupted SF	3488	2025
c) JPEG-corrupted SF	3440	2002
Difference (a)-(b)	110	61
Difference (a)-(c)	160	100

V. CONCLUSIONS AND FUTURE WORK

In this paper, we have presented a novel significant feature extraction method for content-based image authentication applications. This algorithm exploits the redundant contourlet image representation to capture image features across spatial and directional resolutions. The validity and the efficiency of this technique in providing significant image features with a high degree of invariance under acceptable manipulations have been shown through many experiments and comparative studies. Error control coding and watermark techniques are being integrated to enhance the ability of the resulting scheme to reach its most significant aims, namely the authentication of the image's content and the detection of malicious manipulations.

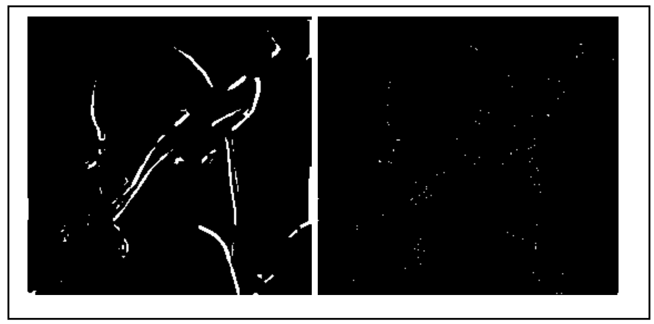

Fig. 3 (left) Significant features for the JPEG-compressed Lena image (0.6 bits/pixel, PSNR=36.3) and (right) the difference with the original image's significant features.

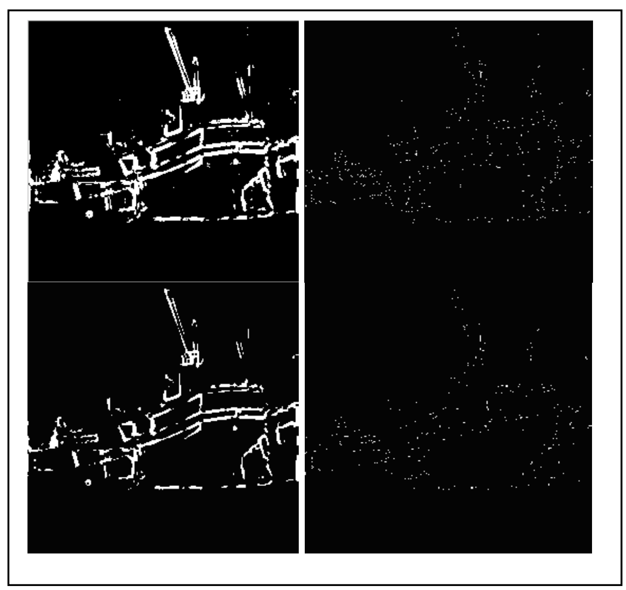

Fig. 4 (Top) Significant features for the JPEG-compressed Boat image (0.8 bit/pixel, PSNR=33.5 dB) and the difference with the original image's significant features. (Bottom) The use of morphological operations reduces the number of map coefficients and enhances the stability of feature values.

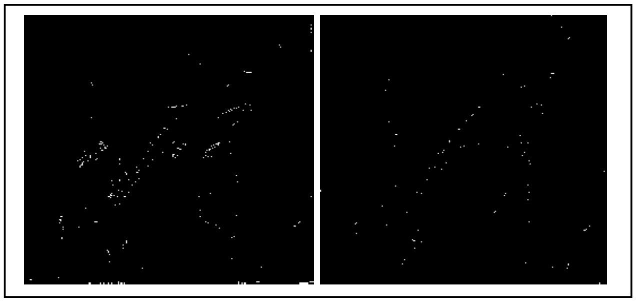

Fig. 5 Comparison between (left) a wavelet-based SF extraction and (right) a contourlet-based SF extraction corresponding to Lena image and Noise-corrupted Lena image (39.6 dB). For each case, a difference between original and corrupted SF is represented. The corresponding map densities are (left) 266 and (right) 61.

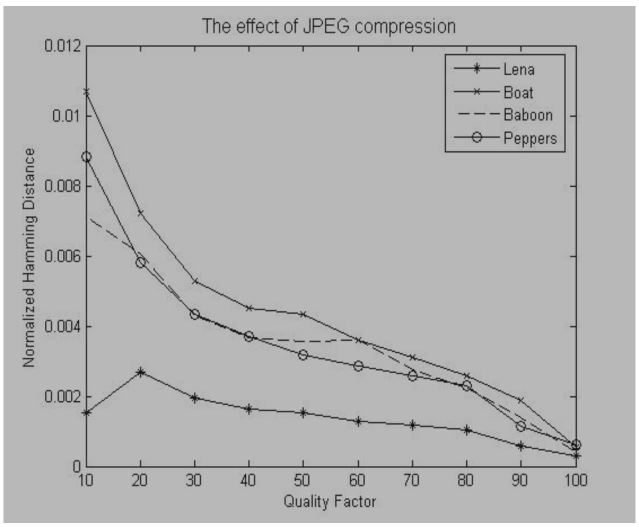

Fig. 6 Stability performance of extracted significant features under JPEG compression.

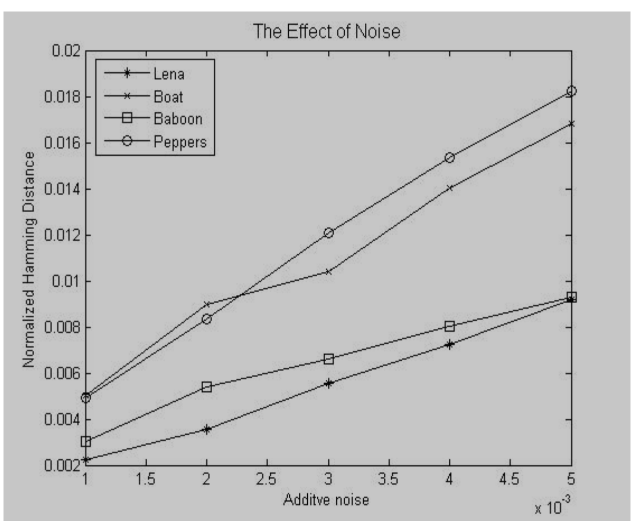

Fig. 7 Stability performance of the extracted significant features under a variety of additive Gaussian noise (x values represent noise variance).

TABLE II. COMPARISON OF MAP DENSITY AND SIGNIFICANT FEATURE (SF) STABILITY BETWEEN A REDUNDANT CONTOURLET –BASED APPROACH AND A REDUNDANT WAVELET-BASED APPROACH. APPLIED MANIPULATIONS CORRESPOND TO ADDITIVE NOISE-39.6DB, AND JPEG-36.3DB.

Image: Lena	**Redundant Contourlet Approach**	**Redundant Wavelet Approach**
a) Original SF	2016	2016
b) Noise-corrupted SF	2025	1826
c) JPEG-corrupted SF	2002	2049
Difference (a)-(b)	61	266
Difference (a)-(c)	100	237

Fig. 8 (left) An example of malicious manipulation on Lena image [http://www.ws.binghamton.edu/fridrich/Research/isspa99_paper3] and (right) significant feature difference with the original image. The manipulated regions can be localized through the cxhange and the increase of the map density.

REFERENCES

[1] N. Baaziz, "Adaptive watermarking schemes based on a redundant contourlet transform," *Proc. ICIP,* Genoa, Italy, September 2005.

[2] M. Barni, F. Bartolini, and A. Piva, "Improved wavelet based watermarking through pixel-wise masking," *IEEE Trans. Image Proc*., vol.. 10, no. 5, May 2001, pp. 783–791.

[3] M. Do and M. Vetterli, "Contourlets: a directional multiresolution image representation," *Proc. ICIP*, Rochester, New York, USA, September 2002, pp. 357–360.

[4] C. Lin and S. Chang, "A robust image authentication method distinguishing JPEG compression from malicious manipulation," *IEEE Trans. On Circuits and Systems of Video Technology*, vol.. 11, no. 2, February 2001, pp. 153–168.

[5] J. Fridrich and M. Goljan, "Robust hash functions for digital watermarking," *Proc. International Conference on Information Technology: Coding and Computing,* March 2000.

[6] M.K. Mihcak and R. Venkatesan , "New iterative geometric methods for robust perceptual image hashing," *Proc. Of ACM Workshop on Security & Privacy in Digital Rights Management,* November 2001.

[7] C. Rey and J.L. Dugelay, " A survey of watermarking algorithms for image authentication," *EURASIP Journal on Applied Signal Processing,* June 2002, pp. 613–621.

[8] Q.B. Sun, S.F. Chang, M. Kurato and M. Suto., "A new semi-fragile image authentication framework combining ECC and PKI infrastructure", *ISCAS02*, Phoenix, USA, May 2002.

[9] Q.B. Sun and S.F. Chang, "Semi-fragile image authentication using generic wavelet domain features and ECC," Proc. *ICIP*, Rochester, USA, September 2002.

[10] A. Swaminathan, Y. Mao and M. Wu, , "Image hashing resilient to geomeric and filtering operations," *Proc. Of IEEE Workshop on Multimedia Signal processing ,* September 2004.

[11] R. Venkatesan, S.M. Koon, M.H. Jakubowski and P. Moulin, "Robust image hashing," ," *Proc. ICIP*, September 2000, pp. 664–666.